

APPENDIX 6.7

Empirical Methods and Sources

National accounts¹ distinguish between the gross value added of domestic businesses operating at home and abroad from all businesses operating within a country (whether domestically or foreign owned). The former is called Gross National Product (GNP) and the latter Gross Domestic Product (GDP) (see appendix table 6.7.1). For instance, US GNP would cover all Ford plants operating in the United States and abroad, while GDP would cover both Ford and Toyota plants operating in the United States. The former might be relevant to the stock market valuation of Ford, while the latter would be relevant to the analysis of domestic employment and capital stock in the US-based auto industry. To go from GNP to GDP, we would have to subtract the gross value added of Ford plants operating abroad and add the gross value added of Toyota plants operating in the United States.

I. Aggregate Operating Surpluses

We are primarily concerned with GDP and its components because available employment and capital stock measures refer to the domestic economy. The next step is to break GDP down into its relevant components. National accounts typically measure aggregate output (gross value added) and corresponding aggregate income (wages plus taxes plus gross operating surplus) separately. The two sides should match in principle, but do not in practice. Since the product measure is considered more reliable, the two sides are reconciled by adding the difference between the two, called the Statistical Discrepancy (SD), to the income side. In the NIPA definition, $NOS = GDP - [SD + \text{compensation of employees within the country} + \text{net indirect business taxes} - \text{economic depreciation, i.e., the depreciation of capital goods valued at current cost, called Consumption of Fixed Capital (CFC)}]^2$. Notice that this procedure implicitly allocates the income side measurement error to the sum of employee compensation and/or net taxes, rather than to Net Operating Surplus (NOS). Appendix table 6.7.2 illustrates these calculations.

While the preceding calculation of net operating surplus is standard, it has three major deficiencies as a measure of the NOS of the *for-profit business* sector which is my concern: it fails to remove nonprofit sectors such as the government, nonprofits serving households, and households; it fails to account for an important asymmetry in the NIPA treatment of corporate and non-corporate businesses; and it fails to adjust for the effects arising from NIPA's inclusion of fictitious (imputed) interest flows.

¹ All BEA data used in this book comes from tables last downloaded in 2011 and may therefore differ from more recent tables. The original downloads can be provided on request.

² BEA (2006, table 1).

Appendix Table 6.7.1 GDP versus GNP, United States, 2009

	2009	Source	Line
Gross national product	14117.2	NIPA Table 1.7.5	4
<i>Less:</i> Income receipts from the rest of the world	642.4	NIPA Table 1.7.5	2
<i>Plus:</i> Income payments to the rest of the world	498.9	NIPA Table 1.7.5	3
Gross domestic product	13,973.7	NIPA Table 1.7.5	1

Appendix Table 6.7.2 Derivation of Domestic Gross and Net Operating Surplus

	2009	Source	Line
Gross Domestic Product	13,973.7	NIPA Table 1.7.5	1
<i>Less:</i> Statistical discrepancy	118.3	NIPA Table 1.7.5	15
Gross Domestic Income	13,855.4	NIPA Table 1.10	1
<i>Less:</i> Domestic compensation of employees (paid)	7,807.2	NIPA Table 1.10	2
<i>Less:</i> Taxes on production and imports less subsidies	963.5	NIPA Table 1.10	9–10
<i>Less:</i> Consumption of fixed capital	1,866.3	NIPA Table 1.10	23
Aggregate Domestic Net Operating Surplus	3,218.4	NIPA Table 1.10	11
Net interest and miscellaneous payments, domestic industries	841.9	NIPA Table 1.10	13
Business current transfer payments (net)	133.4	NIPA Table 1.10	14
Proprietors' income with inventory valuation and capital consumption adjustments	979.4	NIPA Table 1.10	15
Rental income of persons with capital consumption adjustment	289.7	NIPA Table 1.10	16
Corporate profits with inventory valuation and capital consumption adjustments, domestic industries	989.5	NIPA Table 1.10	17
Current surplus of government enterprises	–15.6	NIPA Table 1.10	22

II. Business Sector Accounts

GDP, and hence GOS and NOS, includes nonprofit institutions serving households (NPISH), the so-called household sector (HH), and government. NPISH includes religious, welfare, social service, grant-making foundations, political organizations, museums, and libraries; some civic and fraternal organizations; medical care facilities, educational and research organizations; recreational, cultural, civic and fraternal organizations; and labor unions, legal aid, and professional organizations. However, it does not include nonprofit organizations such as chambers of commerce, trade associations, and homeowner's associations because these are considered to serve businesses. Moreover, the definition of "business" itself includes tax-exempt cooperatives,

credit unions, mutual financial institutions, and university presses (Mead, McCully, and Reinsdorf 2003, 13). Removing NPISHs is therefore an important, albeit not complete, correction. The “household” also includes persons who rent out part of their houses, which we can treat as a nonprofit activity; and an entirely fictitious owner-occupied housing sector (OOH) that arises because NIPA treats private homeowners as “businesses” renting their own homes to themselves. The rationale for the OOH is that the GDP component of “housing services” should be independent of whether a house is lived in by its owner or rented out. Thus, if you live in a home which you owned, you will be treated as a business renting your home to yourself. Even though you live in your home, the rental revenue that you would receive if you did rent it out is treated as the revenue of your owner-occupied housing (OOH) business. The household sector GOS is derived by subtracting estimates of homeowner maintenance, repairs, employee compensation, property taxes, and mortgage interest from the actual and fictitious rental revenue of houses owned by persons. The resulting net income falls within aggregate GOS and NOS under the sub-categories of “rental income of persons” and “net interest paid by businesses” (Ritter 2000, 18n17; Mayerhauser and Reinsdorf 2007, 1).³ NIPA defines the “business” sector as exclusive of NPISH, HH, and general government, but the Bureau of Labor Statistics (BLS) also correctly excludes government enterprises because they are nonprofit (Harper, Moulton, Rosenthal, and Wasshausen 2008, 1 and table A.1). Appendix table 6.7.3 displays the corrected measures for business NOS, which is 79% of the corresponding aggregate value.

Appendix Table 6.7.3 GOS and NOS of the Business Sector (Aggregate – HH – NPISH – GOV)

	2009
Business Gross Value Added	10,189.6
Less: Statistical discrepancy	118.3
Less: Domestic compensation of business employees	5,471.0
Less: Taxes on production and imports less subsidies	812.4
Less: Consumption of fixed capital	1,254.1
Equals: Business Net Operating Surplus	2,533.8
Net interest and miscellaneous payment	343.3
Current transfer payments (net)	127.7
Proprietor’s income with inventory valuation and capital consumption adjustments	979.4
Rental income of persons with capital consumption adjustment	93.9
Corp Profit with inventory valuation and capital consumption adjustments, domestic industries	989.5

Source: Appendix Table 6.8A.1

³ The gross output of OOH is defined as space rent. Materials and repairs are subtracted to get gross value added of OOH, and taxes net of subsidies (there being no employee compensation associated with OOH) to get GOS_{OOH} . The latter is then split into net interest paid (mortgage interest paid minus interest received by OOH business), transfer payments, depreciation, and “rental income of persons” (NIPA Table 7.12, lines 133–140)

III. Wage Equivalent and Non-Corporate and Corporate Rates of Profit

A second problem is that the NIPA measure of business flows embodies a curious asymmetry. Consider two firms, one a corporation and the other an unincorporated enterprise (proprietorship or partnership), each with a value added of \$100 million out of which each pays \$50 million to regular employees and charges \$10 million as depreciation, leaving \$40 million in each coffer. In the case of the corporation, NIPA accounts would record a further \$10 million in salaries, bonuses, and certain stock option exercises by corporate officers as part of corporate employee compensation,⁴ so that the corporate business NOS would be \$30 million. But in the case of the unincorporated business, NIPA would record all of the value added in excess of regular employee compensation (i.e., all \$40 million) as personal earnings of proprietors and partners—officially designated as part of Domestic Personal Income under the designation “proprietors’ income” (Seskin and Parker 1998, M-8). No allowance would be made for the fact that some part of this sum is really the profit of unincorporated enterprises.⁵ Indeed, on this accounting, the profit of the unincorporated sector would be zero. Correcting for this would require us to split proprietors’ income into two parts: a wage-equivalent and a corresponding profit. One way to do this is to assign the average employee compensation of full-time equivalent employees to proprietors and partners (“self-employed persons”⁶) (Shaikh and Tonak 1994, 112–113, 304–305; Gomme and Rupert 2004, 4–5), premised on the assumption that proprietors and partners earn at least as much as private full-time employees. This generates the wage-equivalent WEQ1. However, using the average private wage rate per worker does not capture the rapidly changing ratio of officers’ salaries to wages since the 1970s. An alternate procedure, which is used by the US Bureau of Labor Statistics (BLS) itself, would be to split proprietors’ income into wages and profit in the same proportions as the corporate sector (Jorgenson and Landefeld 2004, 15). Corporate employee compensation includes the salaries of corporate officers, and corporate profit does not. But non-corporate employee compensation does not include the compensation of proprietors and partners, and proprietors’ income implicitly does. To make the sectoral measures consistent, we need to add the salaries of proprietors and partners to the wage bill for their employees and subtract these salaries from proprietors’ income to get non-corporate profit. Assuming that the full wage/profit share is the same in the

⁴ Bureau of Economic Analysis, Chapter 10: Compensation of Employees, <http://bea.gov/national/pdf/ch10%20compensation%20for%20posting.pdf>.

⁵ “The NIPA’s present a single estimate for proprietor’s income with no decomposition of the return to the proprietor for his or her labor and the return to the capital invested in the business. . . . The difficulties with developing such a breakdown are twofold. First, proprietors do not breakdown their income and report the total amount as business income to the tax and statistical authorities. Second, indirect estimates that apply average wages to estimates of hours worked by self-employed persons or capital returns to estimates of capital stocks employed by proprietors result in either negative returns to capital or labor depending upon which imputation is estimated first. The reasons for this are not clear, but may be related to the extent to which proprietors underreport income to tax and statistical authorities, problems in measuring hours worked and capital invested by the self-employed, and the non-pecuniary benefits of self-employment. Better data on proprietor income will have to await improvements in the reporting of self-employment income and hours” (Jorgenson and Landefeld 2004, 15).

⁶ Self-employed persons refers to proprietors and partners (NIPA table 6.7, n. 1).

Appendix Table 6.7.4 Effects of Non-Corporate Wage Equivalents on Profits

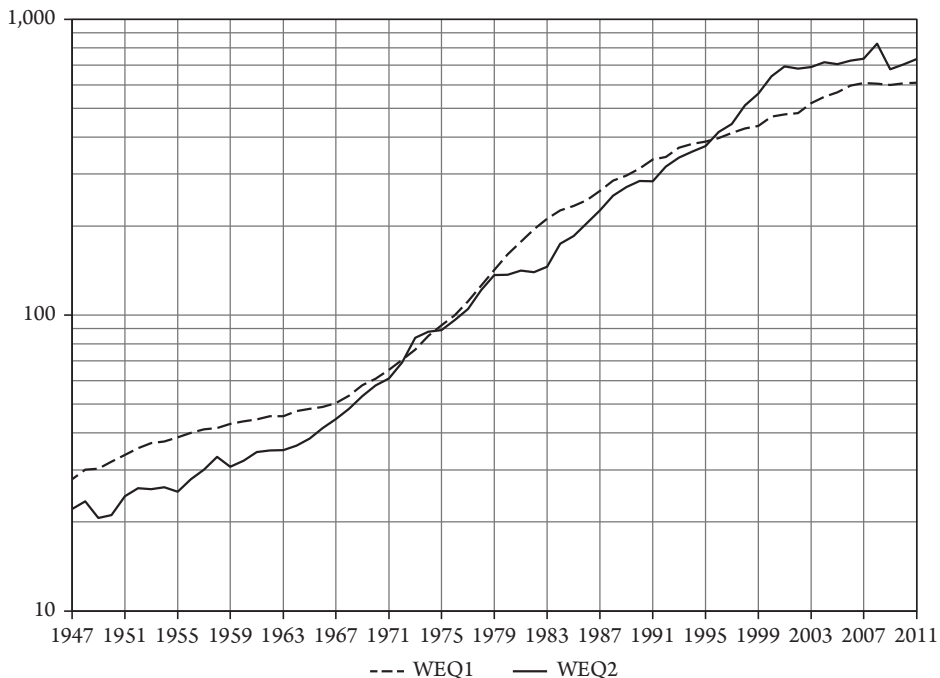
<i>Variable</i>	<i>Source</i>	<i>Variable</i>	<i>2009</i>
Proprietorships and Partnerships Income w/IVA and CCAdj	T 1.13, line 23	PropInc	979.4
Employee Compensation, Proprietorships and Partnerships	ECbusnipa (Appendix Table 7.1A) - ECcorpnipa (T 1.14, line 4)	ECpriv	761.4
Number of Proprietors and Partners (Self-Employed Persons)	T 6.7, line 1 (Thousands-\$)	SEP	9,829
Private Sector Employee Compensation per Full-Time Equivalent Employee (FEE)	T 6.2, line 3/Table 6.3, line 3 (\$)	ecpriv	60,920.9
<i>Estimated Wage Equivalent of Prop & Partner's Income using SEP x Average Private EC per FTE</i>	ecpriv* SEP	WEQ1	598.8
Non-corporate Profit Estimate 1	PropInc - WEQ1	pnoncorpl	380.6
Corporate Wage-Profit Ratio	T1.14, line 4 ÷ line 11		4.76
Employee Compensation, Proprietorships and Partnerships	Ecbusnipa (Appendix 7.1 Profbig)-Eccorpnipa (NIPA Table 1.14, 4)	ECprop	761.4
<i>Estimated Wage Equivalent of Prop & Partner's Income using Corp Wage/Profit ratio</i>	$(\sigma \cdot \text{PropInc} - \text{ECprop}) / (1 + \sigma)$	WEQ2	677.2
Non-corporate Profit Estimate 2	PropInc - WEQ2	Pnoncorp	302.2
Sectoral Profit Rates			
Business Net Fixed Capital, Current-Cost (end of year)	KNCcorp + KNCnoncorp (see below)	KNCbus	16,343.4
Corporate Net Fixed Capital, Current-Cost (end of year)	Fixed Asset Table 6.1, line 9	KNCcorp	12,701.7
Proprietor and Partner Net Fixed Capital, Current-Cost (end of year)	Fixed Asset Table 6.1, lines 6 + 7	KNCnoncorp	3,641.7
Business profit rate	Pbus/KNCbus(-1)	rbus	7.7%
Corporate profit rate	Pcorp/KNCcorp(-1)	rcorp	7.5%
Non-corporate profit rate (using WEQ2)	Pnoncorp/KNCprop(-1)	moncorp	8.1%
Non-corporate profit rate using WEQ1	Pnoncorpl/KNCprop(-1)	rnoncorpl	10.2%

Source: Tables 6.8.I.1-2.

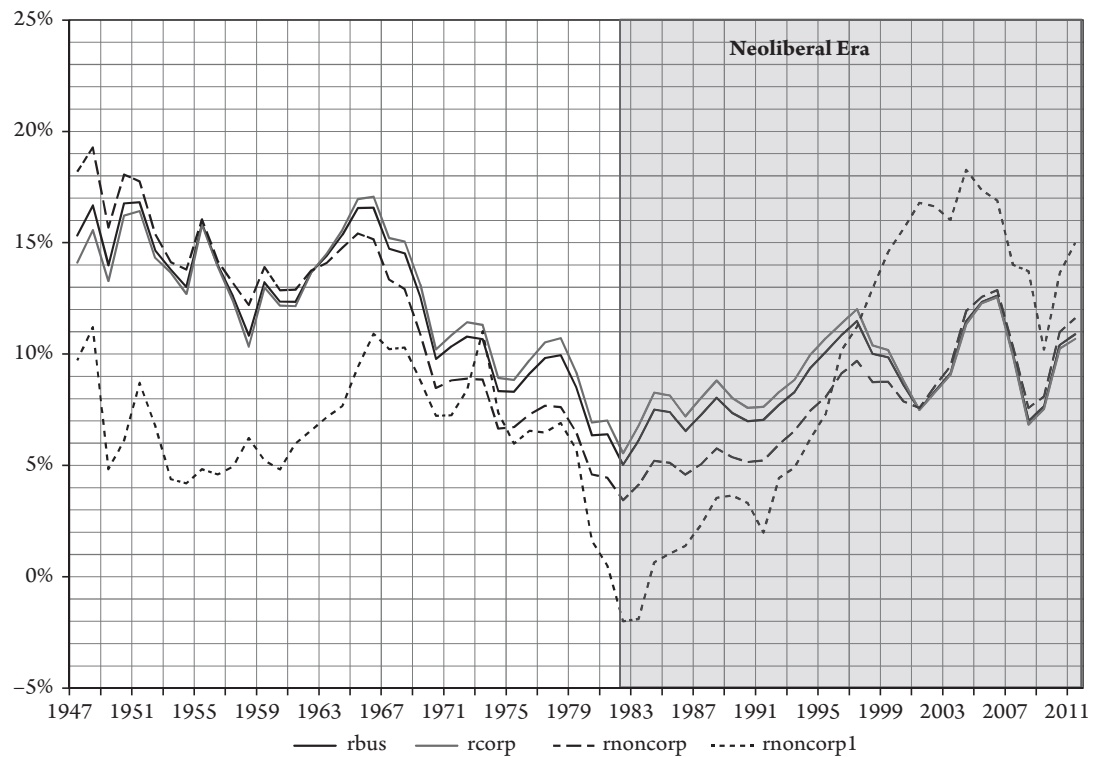
two sectors then allows us to estimate the wage-equivalent (WEQ2) which would make it so (see appendix table 6.8A.2).

Appendix table 6.7.4 illustrates the calculation of these two types of wage-equivalent measures. Appendix figure 6.7.1 plots them over 1947 to 2009. WEQ2 is initially lower than WEQ1, but rises faster. This is not surprising. The former assigns the average private employee compensation to each proprietor and partner (self-employed person), whereas the latter allows for some impact of the rising ratio of salaries to wages from the 1990s onward (Mishel 2006).

Also shown in appendix table 6.7.4 is the impact on non-corporate profits and rates of profit. Appendix figure 6.7.2 displays two non-corporate rates of profit, *mnoncorp1* and *mnoncorp*, corresponding to WEQ1 and WEQ2, respectively, along with the corporate profit rate and that of the overall business sector. WEQ1 yields a profit rate which is less than half the corporate rate for 1947 to 1990, after which it rises rapidly to become more than double the corporate rate at points in the 1990s. WEQ2, on the other hand, yields a much more plausible result. Its corresponding non-corporate rate of profit tracks the corporate rate quite closely, but is a bit smaller. Some part of the size difference may be due to the fact that the estimated non-corporate sector still includes a number of nonprofit enterprises such as chambers of commerce, trade associations, homeowner's associations, tax-exempt cooperatives, and so on, whose very low or even negative "rates of profit" pull down the average non-corporate rate. These sectors are difficult to remove due to lack of adequate data. I use WEQ2 in all subsequent calculations. Hence, the overall business rate of profit shown in appendix table 6.7.4 and appendix figure 6.7.2 is defined as the sum of corporate profit and the non-corporate profit corresponding to WEQ2, divided by the sum of corporate and non-corporate current-cost capital stocks. Similarly, WEQ2 is the basis for the non-corporate components of the corrected business sector derived in appendix table 6.8.I.3).



Appendix Figure 6.7.1 Wage Equivalent Component of Proprietors' Income



Appendix Figure 6.7.2 Effect of Wage Equivalent Adjustment on Sectoral Profit Rates

IV. Actual Versus Imputed Net Interest

The estimation of non-corporate wage-equivalent and profit is not the end of the story because we still have to deal with a third problem arising from NIPA's treatment of interest payments. This does not affect measures of profit which are net of interest payments, but it does affect measures of business GVA, VA, and NOS. For ordinary NIPA sectors, net "interest payments are generally treated as a distribution of income by businesses to investors who have provided them with funds, not as a payment for services" (Fixler, Reinsdorf, and Villones 2010, 347). The trouble is that financial intermediaries such as banks receive more interest than they pay, since that is how they make their money. This means that their net interest "payments" are negative—so much so that a conventional treatment of their value added (the sum of employee compensation, taxes, net interest paid and profits) tends to yield a negative number. Net operating surplus, which in this case would be equal to profits and net interest paid, then also tends to be negative. The obvious solution is to recognize that banks are, after all, financial intermediaries who take a cut out of the money flows passing through their hands (Shaikh and Tonak 1994, 260–267). Indeed, something like this was done until recently by many European countries (Fixler, Reinsdorf, and Villones 2010, 347n7). However, NIPA insists on treating banks as if they were ordinary businesses. And since ordinary businesses must have positive value added and positive gross surplus, "an imputation for implicit financial services produced by banks is included in the NIPA" under the category of net interest paid (Fixler, Reinsdorf, and Villones 2010, 347). Since banks are corporations, corporate net interest ends up being largely "composed of imputations" (Ritter 2000, 18). For instance, in 2009 total net interest within aggregate NIPA NOS is \$841.9 billion, of which \$747.6 billion is imputed (NIPA table 7.12, lines 43, 44).

In the normal course of events, non-financial businesses, households, and so on pay a certain amount of money as net interest to banks. This money constitutes the net revenue of banks (interest received on loans minus interest paid on deposits) and is partitioned into bank costs and profit. In order to accomplish its desired transformation of banking accounts, NIPA makes two interventions to these foregoing accounts: it creates certain imputed quantities to be used in the accounts of non-financial business, household, and so on; and it creates a different imputed quantity to be used in bank accounts. On the side of non-financial business, NOS is reduced by this imputed quantity and total interest paid is reduced by the same amount by adding a negative item called imputed net interest paid. Since by definition $\text{Profit} = \text{NOS} - \text{Total Net Interest Paid}$, the reductions in the latter two cancel out so non-financial profit is unchanged. On the side of banks, the second quantity is subtracted from bank NOS and is simultaneously listed as a negative item called bank total net interest paid, so bank profits are also unchanged. We can then see two things: that the adjustments reduce the NIPA measures of nonfinancial and bank NOS's; and that we can undo these effects by removing the two imputed quantities from NIPA accounts, thereby restoring non-financial and bank NOS's to their proper (classical) levels. In effect, we need to increase non-financial NOS by adding the magnitude of non-financial imputed net interest, and to increase bank NOS by adding the magnitude of bank total net interest paid. Readers who wish to skip the rather detailed presentation that follows may wish to go to the final appendix tables 6.7.10 and 6.7.11.

To understand the exact procedures and their rationale, it is useful to elaborate on the examples in chapter 6, sections II and V, so as to highlight the differences between the business and NIPA treatments of net interest, operating in two steps: interest payment from the production sector to banking (finance) sector; and interest payments from the household sector to

the banking sector. In the first case, suppose the production sector has an net operating surplus (NOS) of 400 out of which it pays 70 as net interest to banks (80 interest paid on its loans minus 10 as interest received on its bank deposits), leaving 330 as production profit. On the side of banks, 70 is received as net interest on loans⁷ from which are subtracted total banking costs of 40 (intermediate input costs of 14, depreciation of 16, and wages of 10), leaving a banking sector profit of 30. Within the classical accounting depicted in appendix table 6.7.5, the production surplus of 400 has been split into production profit (330) and net interest paid (70), a portion of the latter then showing up as additional banking profit (30).

NIPA insists on treating banks in the same manner as the production sector, in which case the net monetary interest received by banks (70) would have to be recorded as a component of NOS in the form of negative net monetary interest paid (−70). To balance out the banking sector accounts in such way as to leave banking profits unchanged, NIPA therefore adds various imputed receipts and payments. Since any imputed payment (receipt) by banks must show up as a receipt (payment) elsewhere in NIPA accounts, in the present case the imputations also show up in the production sector—once again in such a manner as to leave production profits unchanged. The steps involved are, to put it mildly, somewhat complicated (Moulton and Seskin 2003; Fixler, Reinsdorf, and Villones 2010).⁸

The first step is to argue that since the interest rate charged by bank on their loans is higher than that charged in the money market (the reference rate), borrowers must be getting some hidden services from banks for which they are willing to pay. Notice that this empirical accounting procedure relies on the theoretical assumption that every market price is supremely correct and efficient (no monopoly power or asymmetric information here!). On this basis, the monetary interest paid by production (80) is split into smaller interest that would have been paid

Appendix Table 6.7.5 Classical Accounts, Net Interest Paid by Production

	<i>Production</i>	<i>Banks</i>	<i>Total Economy</i>
Total Circulation	1,400	70 { 80	1, 480
Intermediate Financial Input		10	10
Intermediate Commodity Input	500	14	514
GVA	900	56	956
CFC	200	16	216
VA	700	40	740
Wages	300	10	310
<i>Net Operating Surplus</i>	400	30	430
<i>Net Interest Paid</i>	70		70
Monetary Net Interest Paid	70 = 80 − 10		
Imputed Net Interest Paid	—		
<i>NIPA Profit</i>	330	30	360

⁷ For a non-financial business, interest payments are disbursements from its operating surplus. On the other hand, for a bank, the interest paid on its intake (deposits) is part of its operating costs. This distinction will play an important role in the theory of the interest rate developed in chapter 10, section II.

⁸ I am most grateful to Gennaro Zezza for this help in sorting out the mysteries of interest imputations.

at the reference rate (32) and imputed payments for implicit “borrower services” provided by banks (48). The second step is to argue that since the interest rates banks pay on deposits is lower than the reference rate, the total deposit interest received by production (10) must be split into the larger interest that would have been received at the reference rate (15) and the implicit “depositor services” provided by banks without charge, the latter to be recorded as an implicit payment by production and hence a negative receipt (−5). The third step is to replace the actual monetary interest paid (80) and received (10) by the production sector by the corresponding reference rate amounts paid (32) and received (15). Note that the NIPA decomposition of production sector interest payments implies that the reference interest paid by production (32) = the monetary interest actually paid (80) minus the implicit payment for “borrower services” (48), so the replacement can be accomplished by adding in a negative imputed interest paid (−48). In a similar vein, the reference interest supposedly received by the production sector on its deposits (15) = monetary interest received (10) plus imputed interest received in the form of free depositor services (−(−5) = 5), adding the former sum as imputed interest received serves to accomplish the necessary replacement. The net effect is to supplement the existing net monetary interest paid by the production sector (70) with a new sum of imputed net interest paid (−53) representing imputed interest paid (−48) minus imputed interest received (5). Then overall net interest paid, monetary and imputed, is 17: net monetary interest (70) plus net imputed interest (−53), as summarized in the production column of appendix table 6.7.6.

Comparing the classical and NIPA treatments in appendix tables 6.7.5 and 6.7.6, respectively, the replacement of the actual production sector net monetary interest paid of 70 (= 80 − 10) by a reference net monetary and imputed interest paid of 17 (= 70 + (−53)) reduces production sector recorded net interest by an amount $A = 53$. This is exactly the total of services supposedly provided by banks without explicit payment in the form of borrower services (48) and depositor services (5). Therefore the fourth step is to also treat these imputed purchases of banking borrower and depositor services as intermediate inputs for production ($A = 53$) and also as corresponding total sales of banks, that is, their gross output (53), shown in the highlighted areas of appendix table 6.7.6. On the side of production, increasing input costs by $A = 53$ lowers GVA, VA, and NOS by that amount, but because net monetary and imputed interest has been lowered by exactly the same amount, production profit is unaffected. On the side of banking, replacing the actual net monetary revenue of banks (80 interest received on loans minus 10 interest paid on deposits) by the imputed sales of banking services (53) reduced the recorded bank GVA, VA, and NOS by the difference, which the amount $B = 17$. But net banking interest paid is now $-B + -17$ because new recorded (monetary and imputed) net interest paid by production shows up as a net interest paid in the bank column of -17 (i.e., a net receipt of 17). The logic of this is that each particular interest payments in the production column must be recorded in the banking column as a receipt, and vice versa so the signs of the totals are reversed. Since banking revenue has been reduced by B and banking net interest reduced by the same amount, the NIPA measure of banking profits remain unaltered (appendix table 6.7.6)

As noted in chapter 6, section II, we must also consider a case in which the household sector pays interest out of its wage and dividend incomes (themselves derived from the production and banking sectors at this level of abstraction). Suppose the household sector pays 20 in interest on its loans and receives 2 on its deposits. From a classical perspective, since this is a net deduction from the circuit of revenue and the corresponding inflow into the banking circuit of capital gives rise to a net increase of 18 into total business revenue which is in turn split into banking costs

Appendix Table 6.7.6 NIPA Accounts, Net Interest Paid by Production

$A \equiv -$ Imputed Net Interest Paid by Production = $-(-53) = 53$

$B \equiv -$ Total Net Interest Paid by Banks to Production = $-(-17) = 17$

Changes from preceding Classical accounts shown in parentheses in cells

	<i>Production</i>	<i>Banks</i>	<i>Total Economy</i>
Total Circulation	1,400	$53 = 70 - B(-B)$	1,453
Intermediate Financial Input	$53 (+A)$		53
Intermediate Commodity Input	500	14	514
GVA	847 $(-A)$	39 $(-B)$	886
CFC	200	16	216
VA	647 $(-A)$	23 $(-B)$	670
Wages	300	10	310
<i>Net Operating Surplus</i>	347 $(-A)$	13 $(-B)$	360
Net Interest Paid	17 $(-4A)$	-17 $(-B)$	0
Monetary Net Interest Paid	70	-70	
Imputed Net Interest Paid	$-53 = -48 - 5 (-A)$	53	
<i>NIPA Profit</i>	330	30	360

of 12 and new bank operating surplus and new profit of 6. Since this comes out of household flows, production sector operating surplus and profit is unchanged. Hence, aggregate NOS and profit are increased by 6. NIPA proceeds by first splitting monetary interest paid by households (20) into reference interest paid (7) and imputed payments for borrower services (13), and splitting monetary interest received by households (2) into reference interest received (3) and implicit payments of depositor services and hence negative imputed interest receipts (-1). In the present case the replacement of actual net interest payments by reference payments takes place in the household accounts, so it does not affect the production sector. Banking sector gross output is now the total "sales" of borrower and depositor services of 14(= 13 + 1) by banks to households, which after costs of 12 yields a new banking NOS of 2. Total interest paid by banks to households consists of net monetary interest paid on deposits of -18, comprised of interest paid on deposits (2) minus interest received on loans (20); and imputed net interest paid of 14, comprised of imputed interest paid in the form of depositor services (1) minus imputed interest received for loans services (-13). Total net monetary and imputed interest paid by banks is therefore now $-4(= 18 + 14)$. The whole procedure changes actual net receipts of banks from 18 (20 monetary interest received minus 2 monetary interest paid) to 14, for net reduction of $C = 4$, which in turn reduces GVA, VA, and NOS by $C = 4$. At the same time net monetary and imputed interest comes out to $-C = -4$, so bank profit = NOS - Net Interest is unchanged. Appendix tables 6.7.7 and 6.7.8 delineate this second case, first in classical and then in NIPA form.

The combined case of interest payments by non-financial business and households is treated for classical accounts in appendix table 6.7.9 and for NIPA accounts in table 6.7.10.

The differences between combined classical accounts in appendix table 6.7.9 and the corresponding NIPA treatment in appendix table 6.7.10 are located in the columns of the latter

Appendix Table 6.7.7 Classical Accounts, Net Interest Paid by Households

	<i>Production</i>	<i>Banks</i>	<i>Total Economy</i>
Total Circulation	1,400	20	1,420
Intermediate Financial Input		18 { 2	2
Intermediate Commodity Input	500	5	505
GVA	900	13	913
CFC	200	3	203
VA	700	10	710
Wages	300	4	304
<i>Net Operating Surplus</i>	400	6	406
Net Interest Paid			
Monetary Net Interest Paid			
Imputed Net Interest Paid			
<i>NIPA Profit</i>	400	6	406

Appendix Table 6.7.8 NIPA Accounts, Net Interest Paid by Households

$C \equiv -\text{Net (Monetary \& Imputed) Interest Paid by Banks to Households} = -(4) = 4$

	<i>Production</i>	<i>Banks</i>	<i>Total Economy</i>
Total Circulation	1,400	14 (–C)	1,414
Intermediate Financial Input			
Intermediate Commodity Input	500	5	504
GVA	900	9 (–C)	909
CFC	200	3	203
VA	700	6 (–C)	706
Wages	300	4	310
<i>Net Operating Surplus</i>	400	2 (–C)	402
Net Interest Paid		–4 (–C)	
Monetary Net Interest Paid		–18 = 2 – 20	
Imputed Net Interest Paid		14 = 1 – (–13)	
<i>NIPA Profit</i>	400	6	406

as changes shown in parentheses. Only two items are needed to account for these differences: non-financial business net imputed interest paid ($-A = -53$); and bank net (monetary and imputed) interest paid ($-B - C = -(-17) - 4 = -21$). Consider the production column first: intermediate inputs have been expanded by $A = 53$, which lowers GVA, VA, GOS, and NOS by the same amount. But the negative of this same amount is introduced as imputed net interest paid, so that profits are unchanged. Next consider the banking column, in which net bank revenues which were previously at 88 in appendix table 6.7.9 (interest received on loans of 100 minus interest paid on deposits of 12) are now at 67 in appendix table 6.7.10, which is a change of $-B - C = -21$. This changes bank GVA, VA, GOS, and NOS by the same amount. However, bank net interest paid was previously zero in appendix table 6.7.9 and is now $-B - C = -21$. Therefore, the reduction in recorded bank NOS is exactly equal to the reduction in its recorded bank net (monetary and imputed) interest,

Appendix Table 6.7.9 Classical Accounts, Production and Household Net Interest

	<i>Production</i>	<i>Banks</i>	<i>Total Economy</i>
Total Circulation	1,400	88 { 100	1,500
Intermediate Financial Input		12	12
Intermediate Commodity Input	500	19	519
GVA	900	69	969
CFC	200	19	219
VA	700	50	750
Wages	300	14	314
<i>Net Operating Surplus</i>	400	36	436
Net Interest Paid	70		70
Monetary Net Interest Paid	70		70
Imputed Net Interest Paid			
NIPA Profit	330	36	366

Appendix Table 6.7.10 NIPA Accounts, Production and Household Net Interest

A $\equiv$ - Imputed Net Interest Paid by Production = $-(-53) = 53$

B + C $\equiv$ - Net Interest Paid by Banks = $-(-17)$ to Production + $-(-4)$ to Households = 21

	<i>Production</i>	<i>Banks</i>	<i>Total Economy</i>	<i>Amount to be added back</i>
(1) Total Circulation	1,400	88-B-C { 67	1,467	B + C
(2) Intermediate Financial Input	53 (+A)	-	53	
(3) Intermediate Commodity Input	500	19	519	
(4) GVA = (1) - (2) - (3)	847 (-A)	48 (-B - C)	895	A + B + C
(5) CFC	200	19	219	
(6) VA = (4) - (5)	647 (-A)	29 (-B - C)	676	A + B + C
(7) Wages	300	14	314	
(8) <i>Net Operating Surplus</i> = (6) - (7)	347 (-A)	15 (-B - C)	362	A + B + C
(9) Net Interest Paid	17 (-A)	-21 (-B - C)	-4	A + B + C
(10) Net Monetary Interest Paid	70	-88 = -70-18		
(11) Net Imputed Interest Paid	-53 (-A)	67 = 53+ 14		
(12) NIPA Profit	330	36	366	0

so bank profit is unchanged. Note that adding back the corrections indicated in the last column of appendix table 6.7.10 recovers the classical (and business) sums: total NOS (436) in appendix table 6.7.9 = NIPA total NOS (362) + the adjustment factor ($-A - B = 74$) in appendix table 6.7.10.⁹

The corresponding empirical calculations are presented in appendix table 6.7.11. The first two rows display the two items needed, and the third row displays the adjustment term. Items (4)–(7) represent NIPA measures of for-profit business, while the next four lines display them after removal of the imputed interest effects. Item (8) is net imputed interest paid by the non-financial corporate sector (the corporate equivalent of item (2) pertaining to all business) and item (9) is the overall corporate imputed interest adjustment, the sum of items (8) and (1) (which is common to the adjustments for all business and corporate business). In 2009 the imputed interest adjustments raise business and corporate net value added measures by only 1% and 2%, respectively, but raise business and corporate net operating surpluses by 7% and 10%, respectively. But non-corporate operating surplus are also lowered by the transfer out of the wage equivalent of proprietors and partners, and this swamps the positive effect of the imputed interest adjustment so that the overall business sector NOS is lowered by more than 30%. Corporate measures do not have this problem, since the corporate sector already excludes nonprofits and has no need of a wage-equivalent adjustment. Here the only effect is that of the restoration of net monetary interest into GOS/NOS, which raises these measures back to their business levels.

There is one last important point about profitability measures. Adjusted net operating surplus includes actual monetary net interest paid on business debt which makes the appropriate foundation for the general rate of profit in the classical sense, the rate that is turbulently equalized by the inter-sectoral mobility of capital. The adjusted NOS is also “conceptually similar to the financial accounting concept of earnings before interest and taxes” (Mead, Moulton, and Petrick 2004, 3–4). This type of measure becomes important when we consider the classical and Keynesian arguments that investment is driven by the difference between the rate of profit and the rate of interest (chapter 16), for that requires us to start with a measure of profit prior to payment of actual net interest (i.e., earnings before interest and taxes (EBIT)) if we are to compare the rate of profit to the interest rate. The corporate measure is particularly apposite because the similarity of non-corporate and corporate rates of profit (appendix figure 6.7.2) makes the latter a good proxy for the overall rate of profit and because it only requires an easily calculated adjustment for imputed interest.

⁹ The recovered accounts are essentially business accounts, so the aggregate net operating surplus is the one actually garnered by aggregate for-profit business. At this level of abstraction, it is also the classical NOS. However, when we treat wholesale/retail trade, classical and business accounts diverge because the former treats the costs of trading as uses of the total surplus, some part of which shows up as the NOS and profits of the trade sector. But we are concerned here with the NOS and profit of aggregate business, which is a different question from that of their ultimate source (Shaikh and Tonak 1994, ch. 3). Since NIPA does not add any imputations for the trade sector, we can treat them as part of the overall non-financial sector. The important point is that NIPA measures of NOS are different from business measures because of interest imputations.

Appendix Table 6.7.11 Impact of Wage Equivalent and Imputed Interest on Business Sector Accounts

<i>Description</i>	<i>Source</i>	<i>Variable</i>	<i>2009</i>
(1) Bank (Financial Corporate) Net Int Paid ^a	T 7.11, lines (4 + 44 +73) – (28 + 52 + 91)	BankNetIntPaid	–37.6
(2) NonFin Business Net Imputed Int Paid ^b	T 7.11, lines (74+75) – lines (53 + 54)	NFNetImpIntPaid	–136.1
(3) – Fin Corp Net Int Paid – NonFin Bus Net Imputed Int Paid	–(1) –(2)	BusImpIntAdj	173.7
<i>Business Measures</i>			
(4) Business GVA NIPA	Appendix table 6.7.3	GVAbusnipa	10,189.6
(5) Business VA NIPA	Appendix table 6.7.3	VAbusnipa	8,935.5
(6) Business Sector GOS NIPA	Appendix table 6.7.3	GOSbusnipa	3,787.9
(7) Business Sector NOS NIPA	Appendix table 6.7.3	NOSbusnipa	2,533.8
Final Business Sector GVA	(4) + BusImpIntAdj	GVAbus	10,363.3 (100.9%)
Final Business Sector VA	(5) + BusImpIntAdj	VAbus	9,109.2 (101.0%)
Final Business Sector GOS	(6) –WEQ2 + BusImpIntAdj	GOSbus	3,284.4 (73.7%)
Final Business Sector NOS	(7) –WEQ2 + BusImpIntAdj	NOSbus	2,030.3 (68.0%)
<i>Corporate Measures</i>			
(8) NonFin Corporate Net Imputed Interest Paid ^c	T 7.11, line 74 – line 53		–95.8
(9) – Fin Corp Mon Int Paid – NonFin Corp Net Imputed Int Paid –	–(1) – (8)	CorpImpIntAdj	133.4
Final Corp GVA	NIPA Corp GVA + CorpImp-IntAdj	GVAcorp	7,793.3 (101.7%)
Final Corp VA	NIPA Corp VA + CorpImp-IntAdj	VAcorp	6,762.9 (102.0%)
Final Corp GOS	NIPA Corp GOS + CorpImp-IntAdj	GOScorp	2,447.2 (105.8%)
Final Corp NOS	NIPA Corp NOS + CorpImp-IntAdj	NOScorp	1,416.8 (110.4%)

^a Corp Financial (Bank) Net Int Paid = (Corp Fin Mon Int Paid + Corp Fin Imp Int Paid + Corp Fin Borrower Services Paid) – (Corp Fin Mon Int Rec + Corp Fin Imp Int Rec + Corp Fin Borrower Services Received) , line numbers in NIPA T 7.11 downloaded 3/14/2011

^b NonFin Business Net Imputed Int Paid = (Non Fin Corp Borrower Services Paid + Prop/Partners Borrower Services Paid) - (Non Fin Corp Imp Int Received + Prop/Partners Imp Int Received)

^c NonFin Corp Net Imp Int Paid = Non Fin Corp Borrower Svces Paid – Non Fin Corp Imp Int Received.

Source: Appendix table 6.8.I.3.

V. Capital Stock

1. Empirical accuracy of generalized perpetual inventory method rules for aggregated chain-weighted stocks

We can now demonstrate the considerable accuracy of the preceding accumulation rules for chain-weighted aggregate capital stocks developed in appendix 6.5, section V. In the case of net stocks, which are the only ones now officially calculated for the United States, the parameter z_t represents the average depreciation rate, in both nominal and real terms. Since individual depreciation allowances can always be written as the product of some individual depreciation rates and corresponding nominal capital stocks,¹⁰ we get

$$\mathfrak{d}_t = \frac{\mathcal{D}_t}{p'_{K_t} KR_{t-1}} = \sum_{i=1}^n \mathfrak{d}_{it} \left(\frac{p'_{it} KR_{it-1}}{p'_{K_t} KR_{t-1}} \right) = \text{the average depreciation rate} \quad (6.7.1)$$

where the weights are the shares of individual reflatd values of past real stocks.

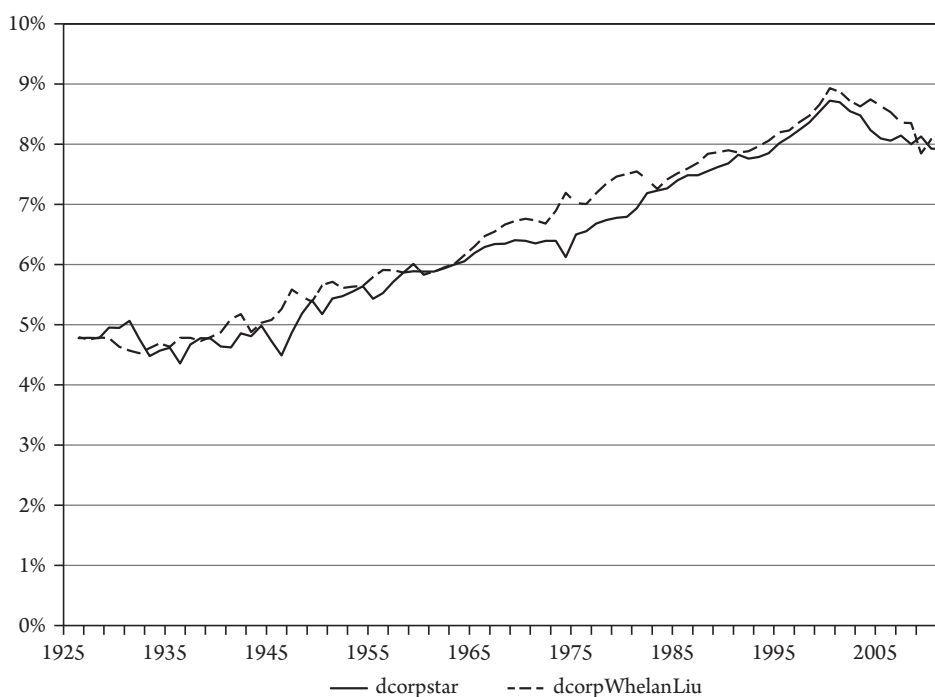
Parenthetically, it is interesting to note that there is no consensus in the literature on the appropriate notion of an aggregate depreciation rate precisely because up to now “no solution has been found” to the problem of an accumulation rule for chain aggregate capital stocks (Liu, Hamalainen, and Wong 2003, 37n12). Various ad hoc rules have therefore been suggested, the most popular being the ratio of nominal depreciation to nominal capital, which is essentially a weighted average of individual depreciation rates with nominal capital stock shares as weights (Whelan 2000, 16; Liu, Hamalainen, and Wong 2003, 37–39):

$$\mathfrak{d}_{WL_t} = \frac{\mathcal{D}_t}{K_{t-1}} = \sum_{i=1}^n \mathfrak{d}_{it} \left(\frac{K_{it-1}}{K_{t-1}} \right) = \text{Whelan – Liu approximate depreciation rate} \quad (6.7.2)$$

But now we have derived the accumulation rule for chain aggregate stocks, we also have the theoretical definition of the depreciation rate. The Whelan–Liu measure is an approximation to the correct rate because its weights are based on past current values ($p'_{it-1} KR_{it-1}$) rather than on reflatd values of past real stocks $p'_{it} KR_{it-1}$. Appendix figure 6.7.3 compares the Whelan–Liu measure to the theoretically appropriate depreciation rate, both for the corporate sector (appendix data table 6.8.II.3).

We test the accuracy of the proposed chain-aggregate accumulation rules by first estimating the depreciation rate through equation (6.5.20) and using it to generate each year’s chain-weighted stock according to equations (6.5.22) and (6.5.23) for current and real stocks, respectively. Recall that the actual aggregate BEA stocks are derived from the chain-aggregation of individual stocks, and that no rule was previously known to characterize the behavior of the corresponding aggregates. In our calculations, each year’s aggregate is derived from the preceding aggregate, using only current gross investment and the aggregate chain-weighted price index of capital. Since the accumulation rule for real stocks is just an algebraic transformation of the current value rule, the error is the same for both. Appendix table 6.7.12 shows that the average error over 1947 to 2005 is a mere one-half of 1%. This establishes that we now have the tools needed to generate alternate measures even for chain-weighted capital stocks.

¹⁰ In practice the choice of a particular depreciation method defines a level of individual real depreciation $\mathcal{D}\mathcal{R}_{it}$ which we can always express, for some time-varying set depreciation rate $\mathfrak{d}_{it}$ as $\mathcal{D}\mathcal{R}_{it} = \mathfrak{d}_{it} KR_{it-1}$. Then nominal depreciation is given by $\mathcal{D}_{it} = p'_{K_{it}} \mathcal{D}\mathcal{R}_{it-1} = \mathfrak{d}_{it} \left(p'_{K_{it}} KR_{it-1} \right)$.



Appendix Figure 6.7.3 Theoretical and Consensus Approximation Depreciation Rates

Appendix Table 6.7.12 Accuracy of Chain-Aggregate Capital Stock Accumulation Rules, US Corporate Fixed Capital, 1925–2009

Average Ratio of Approximate to BEA Current-Cost Stock	99.60%
Average Ratio of Approximate to BEA Constant-Cost Stock	99.60%

Source: Appendix 6.8.II.1.

2. Effects on capital stock measures of alternate assumptions

For reasons discussed in section II of this chapter and in appendix 6.3, the classical measure of the rate of profit is defined in terms of current-cost gross stock. In this case an individual machine is kept on the books at the equivalent of its current market price until it is retired, whereas in current-cost net stock calculations this same sum is reduced by accumulated depreciation. Therefore gross stocks are always larger than net stocks and may have different trends. These differences will be reflected in the corresponding measures of the rate of profit, which could be important when we are comparing the profit rate to the rate of interest, for instance.

In 1997 the BEA adopted the algebraically convenient assumption that all asset efficiencies decline geometrically over an infinite lifetime. This assumption has well-known empirical deficiencies (Harper 1982, 10, 30; Hulten 1990, 125). It also makes it impossible to calculate gross stocks at all because gross stock measures depend on the pattern of retirements, and infinitely lived assets never retire. Hence, the BEA can now only produce measures of net stocks.

Chain-weighting procedures have so far prevented researchers from producing alternate aggregate capital stock measures because there were no known accumulation rules which applied

to chain-aggregates. The derivation of generalized PIM accumulation rules in Appendix 6.5.V.3 removes this limitation. We will address the impact on the estimated path of the postwar capital stock of three types of modifications of the BEA assumptions: the effects of different initial values used in the generalized PIM accumulation rules; the effects of different retirement and depreciation patterns; and the effects of changes in scrapping rates in the face of the Great Depression of 1929–1939. As we will see, it is the latter which has the most dramatic effect on the trends and levels of the postwar capital stocks.

i. Effects of alternate initial values

Appendix table 6.7.13 lists three different 1925 values of current-cost corporate net stock available in various BEA publications (with the latter two also expressed as percentages of the BEA 2011 initial value). Appendix figure 6.7.5 illustrates the effects of alternate initial values (in 1925) for the GPIM accumulation rules, keeping the aggregate depreciation rate the same as in present day net-stock BEA estimates and sticking to the underlying BEA assumption that individual depreciation rates are entirely invariant to economic fluctuations. Only one-third of any initial difference in 1925 remains in 1947: the 31% initial difference of the BEA 1993 starting point is reduced to a 10% difference by 1947, and the 21% initial difference of the SCB 1985 starting point is reduced to 7%. By forty-four years, in 1969, both differences are less than 2%. This indicates that the starting point is not important for the “long run” path of capital stock. But since all starting points end up on the same path in the long run, *the lower initial values must grow faster to catch up with the common long-run path*. In other words, lower starting points yield higher capital stock growth rates for a considerable length of time. This property turns out to have major consequences for the path of the postwar capital stock when we adjust for the effects of the Great Depression (see section II.2.4).

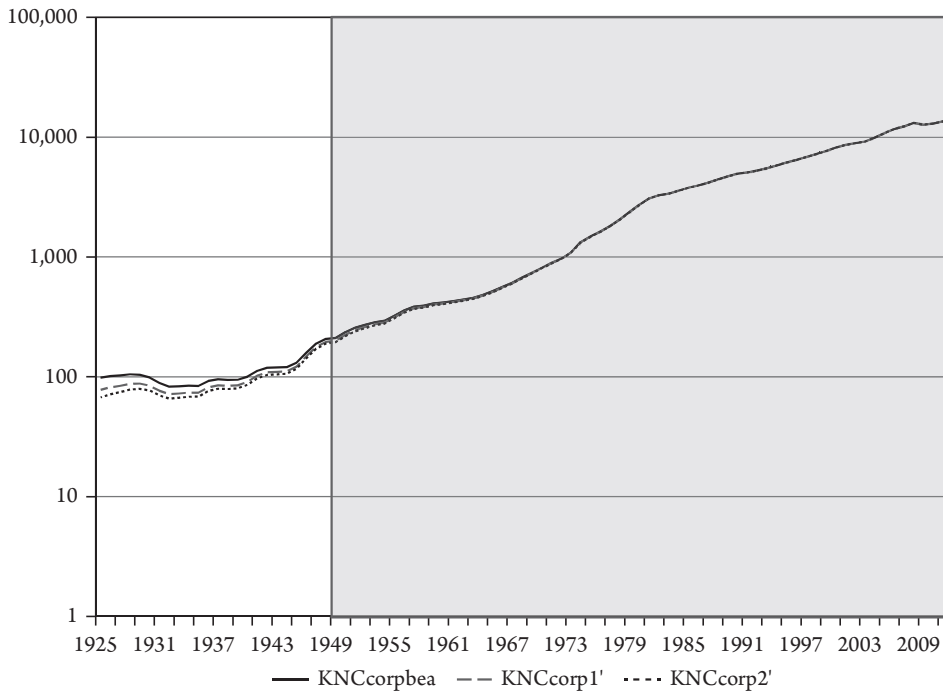
ii. Effects of alternate depreciation and retirement rates

Prior to 1997, the BEA assumed that different types of individual assets had different types of useful lives. This was superior to the current BEA assumption that assets are never scrapped. However, both methodologies suffer from the defect that depletion rates are assumed to be invariant to economic conditions, to which we will return in the next section. The earlier BEA data gives rise to aggregate useful lives and depreciation rates which vary somewhat over time, but since individual useful lives are assumed to be fixed, this aggregate variability is solely due to changes in the asset-mix. We take these earlier aggregate retirement and depreciation rates up to 1997, project them into the present, and use them in equations (6.5.22) and (6.5.23) to

Appendix Table 6.7.13 Different Initial Values in 1925, Current-Cost Corporate Net Stock

<i>Data Source</i>	<i>Location</i>	<i>Value (Bill-\$)</i>
BEA 2011	Fixed Asset Table 6.1, line 1	98.1
BEA 1993	Table A.13, 294 (BEA 1993)	77.7(79%)
SCB 1985	Table 6, 56 (Gorman, Musgrav, Silverstein, and Comins 1985)	67.1(68%)

Source: Appendix Table 6.8B.2.

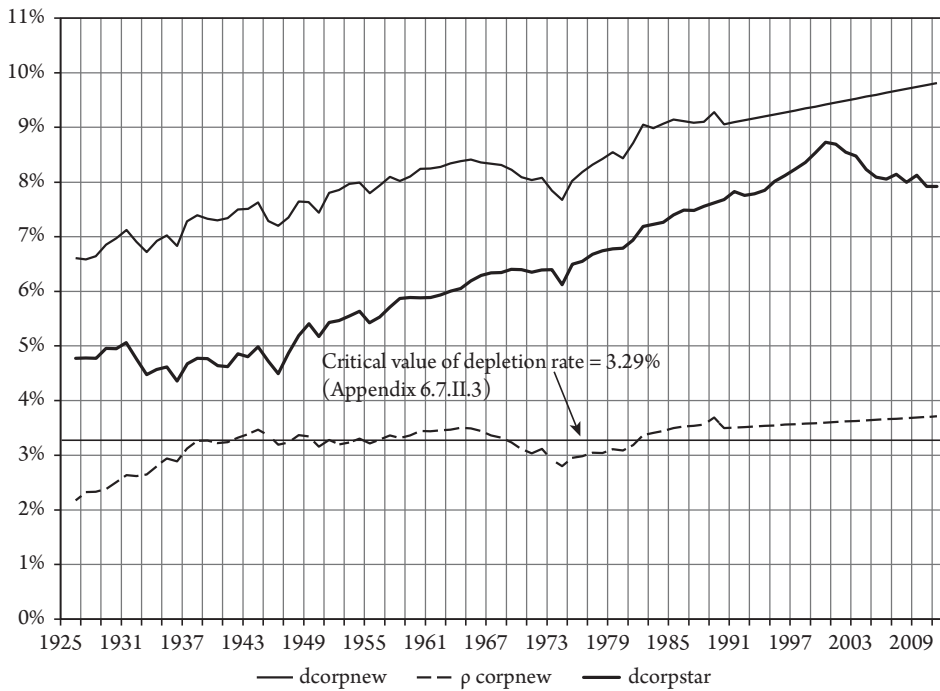


Appendix Figure 6.7.4 Effects of Initial Values on the Path of Capital Stock

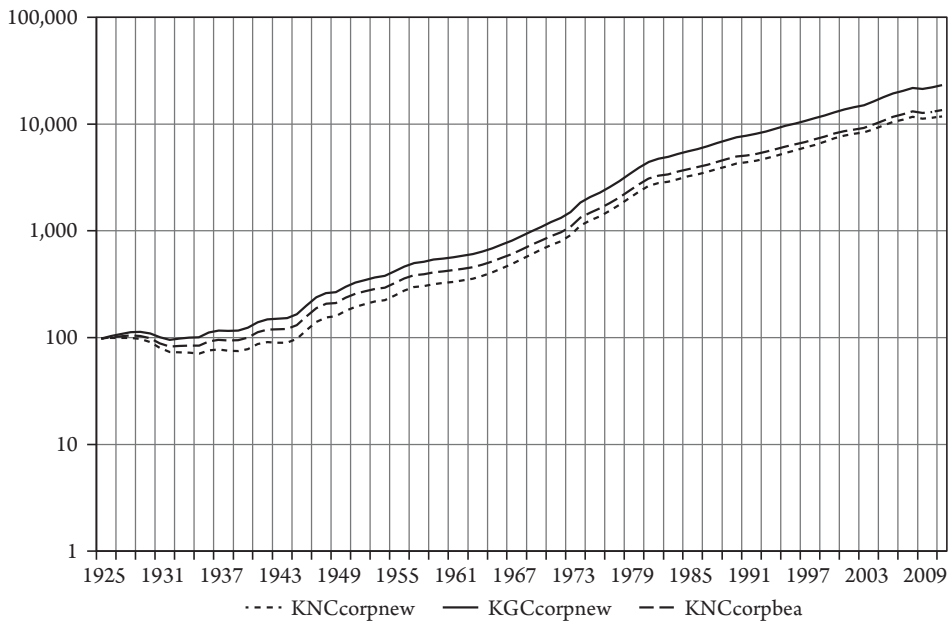
generate new estimates of chain-aggregate gross and net stocks. In order to examine the pure effect of changes in depletion rates, all estimates begin from the initial value used by the BEA in its 2011 data set (appendix table 6.8B2). The starting point for the gross stock estimates is the same as for the net stock because we are interested here in the pure effects of changes in depletion rates. But in the final estimates presented in appendix table 6.8B the gross stock initial point will be adjusted to account for the fact that gross stocks are larger than net stocks.

Appendix figure 6.7.5 compares corporate depreciation and retirement rates derived from the BEA 1993 data to the aggregate corporate depreciation rate implicit in current BEA data. We see that the earlier depreciation rate is higher than the present one. This is essentially because the earlier BEA calculations were based on finite useful lives for fixed assets, whereas the present BEA ones assume infinite useful lives. Of particular note is the fact that the earlier BEA retirement rate is far lower than either of these. It is in fact often lower than a particular critical value also shown on this chart. The significance of this value will be addressed in appendix 6.7, section 4.

Appendix figure 6.7.6 applies the new depreciation rate based on finite service lives to generate new current-cost net capital stock measures for the corporate sector, all beginning from the initial value of the BEA 2011 corporate stock also shown in this figure. The new net stock estimates are generally lower than the current BEA ones, but the relative gap diminishes over time as the two converge. On the other hand, the new gross capital stock are larger than the modern BEA net stock estimates and grow faster over the postwar period. These two different patterns are clear when we look at the ratios of two new estimates to the present-day BEA stocks, as in appendix figure 6.7.7: the relative new net stock measure is fairly stable in the postwar period, while the relative gross stock measure continues to rise throughout. The analytical reason for this difference is addressed in the next section.



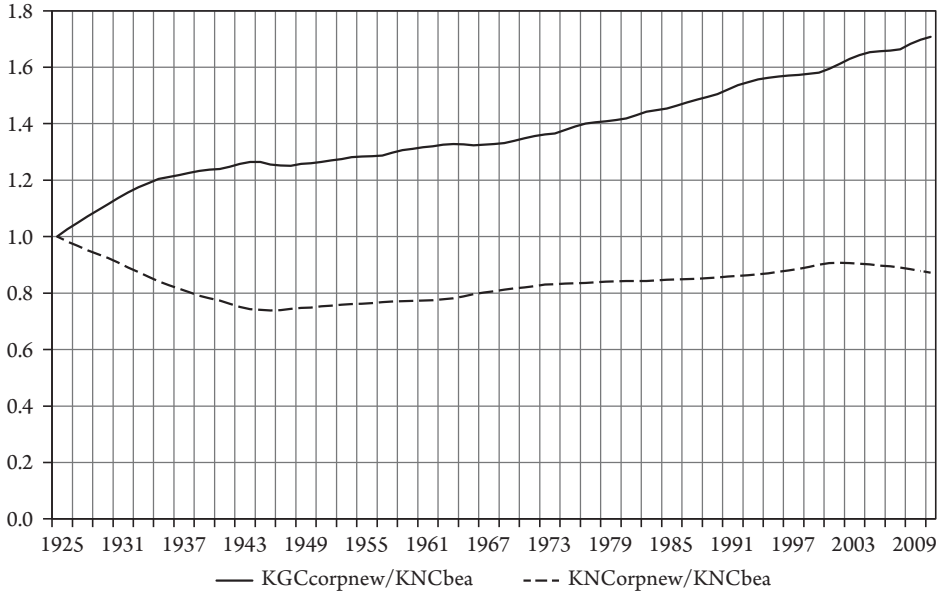
Appendix Figure 6.7.5 Alternate Retirement and Depreciation Rates



Appendix Figure 6.7.6 Effects of Alternate Depletion Rates on the Levels of Capital Stocks

3. Digression on the algebra of the perpetual inventory method

The GPIM rules for current- and constant-cost capital stocks in equations (6.5.22) and (6.5.23) are first order difference equations. Since one can be derived from the other, it is sufficient to consider the properties of equation (6.5.22): $K_t = IG_t + z'_t \cdot K_{t-1}$, where $z'_t \equiv (1 - z_t) \frac{P_{K_t}}{P_{K_{t-1}}}$.



Appendix Figure 6.7.7 Effects of Alternate Depletion Rates on the Relative Levels of Capital Stocks

In actual data, gross investment grows at a roughly constant rate, so we can take $IG_t \approx IG_0 (1 + g_I)^t$, where IG_0 = the initial value of gross investment at time $t = 0$ and g_I = its average growth rate. In addition, at an empirical level $(1 + g'_{PKt}) \equiv \frac{P'_{Kt+1}}{P'_{Kt-1}} \approx \text{constant}$ and at first approximation we can take the depletion rate (z_t) as constant, which make $z' \approx \text{constant}$. The GPIM rule for current-cost stock then corresponds to a first order difference equation with constant coefficients and an exponential forcing term:

$$K_t \approx IG_t + z' \cdot K_{t-1} \quad (6.7.3)$$

where $z' \equiv (1 - z) (1 + g'_{PK})$. Such an equation has the general solution:¹¹

$$K_t = A(z) x^t + C(z) \cdot IG_t \quad (6.7.4)$$

in which both $C(z) \equiv \left(\frac{(1+g_I)}{(1+g_I)-(1-z)(1+g'_{PK})} \right)$ and $A(z) = K_0 - C(z) \cdot IG_0$ are generally positive and constant for any given depletion rate z .

The second term $C(z) \cdot IG_t$ in the general solution rises exponentially because it is proportional to gross investment IG_t . But the first term $A(z) \cdot (z')^t$ is different, because it rises or falls according to whether $z' \equiv (1 - z) (1 + g'_{PK}) \gtrless 1$. *This is where the depletion rate plays a critical*

¹¹ The general solution of $c_1 \cdot y_t + c_0 \cdot y_{t-1} = B \cdot d^t$ is $y_t = A(-b)^t + \left(\frac{d}{c_1 d + c_0} \right) B d^t$ if $c_1 \cdot d + c_0 \neq 0$, where c_0, c_1 are constant coefficients, $b = c_0/c_1$ and A is a constant determined by initial conditions (Gandolfo 1985, 14–20). In the present case $c_1 = 1$, $c_0 = -z' = -(1 - z) (1 + g'_{PK})$, $b = -c_0 = z'$, $B = IG_0$, $d = (1 + g_I)$ so $B \cdot d^t = IG_0 (1 + g_I)^t = IG_t$, and in general $c_1 \cdot d + c_0 = (1 + g_I) - (1 - z) (1 + g'_{PK}) \neq 0$. This yields text equation (6.7.4).

role, because $z' \geq 1$ as $z \geq \frac{g'_{PK}}{(1+g'_{PK})}$. In the United States from 1947 to 2009, the average growth factor for the price of capital goods $(1 + g'_{PK}) \approx 1.034$. From this point of view, any $z > 3.29\%$ would make $z' < 1$,¹² in which case the first term in the general solution will decline continuously, leaving the second term to dominate the long path. Under these conditions, all initial values will converge to the same long-run path. This is the situation depicted in appendix figure 6.7.4 because the BEA 2011 depreciation used for that experiment lies between 5% and 8%. The BEA 1993 net stock depreciation rate is even higher, between 6.5% and 9.5%, so it would yield the same result. But the gross stock retirement rate is a different matter because it varies between 2.2% and 3.7%, so that it is below or near the critical value of 3.29% for most of the time (appendix figure 6.7.5). This, I would argue, is why the relative measure of the new gross current-cost capital stock does not converge to the BEA net capital stock value within the eighty-five-year time span from 1925 to 2009, whereas the relative measure of the new *net* current-stock eventually does (appendix figures 6.7.6 and 6.7.7). The two corresponding “long runs” are different. Of course, empirical rates of depletion and of price change are variable, which makes the actual path somewhat more complicated. Nonetheless, the explanation for the difference in pattern lies in the structural parameters. With this in mind, we are ready for the final experiment, which is to allow for the impact of the Great Depression on the depletion rates of fixed capital.

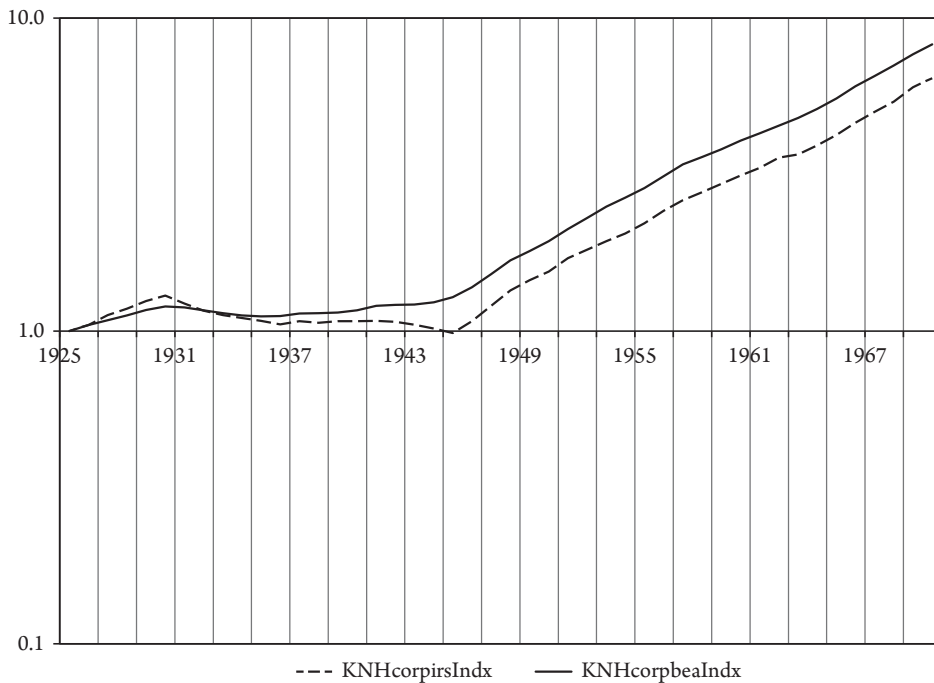
4. Effects on the capital stock of the Great Depression and World War II

Current BEA methodology assumes infinite service lives, which is why the earlier BEA methodology based on (point estimates of) actual useful lives is preferable. But both methodologies suffer from the fact that they assume that depletion rates are invariant to economic conditions. In this section, we examine the consequences of allowing for changes in depletion rates due to the cataclysmic events of the Great Depression and World War II. Here, we make use of the fact that historical stock estimates derived via the PIM are analogous to company book value data on capital stocks.

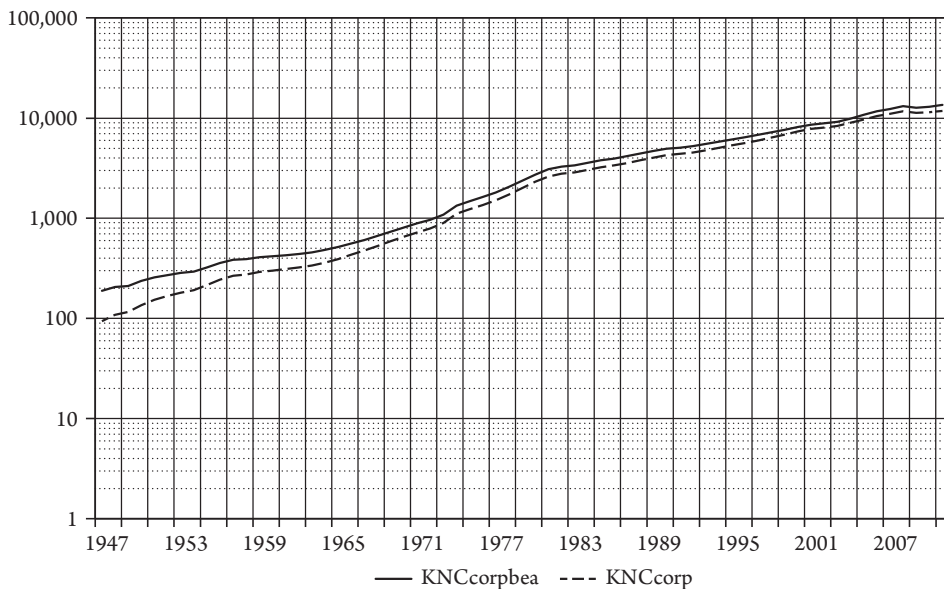
The US Historical Statistics of Income (Census 1975, Series V 115, 924–926) contains data on the book value of net capital assets of all corporations derived from US Internal Revenue Service (IRS) statistics. Net assets as defined there include land, but we can use them to estimate the path of the historical net capital stock. Appendix figure 6.7.8 compares the movements of BEA 2011 historical net stock to that of IRS book value stock, each indexed to 100 in 1925. As one can see, the two behave very differently from 1925 to 1945, but move in similar ways thereafter. By 1947 the IRS book value index has risen by just 20%, whereas the BEA historical capital stock index has risen by 52% (appendix table 6.8.II.4). This difference is not surprising, given that the BEA measure is based on the *assumption* that scrapping of fixed capital is entirely independent of economic conditions.

Since current- and constant-cost capital stocks are based on the same depreciation rates as historical stocks, we can adjust all three for the effects of the interwar period by multiplying them by the IRS book value index from 1925 to 1947 and reverting to the GPIM calculation in equations (6.5.22) and (6.5.23) thereafter. This would bring the BEA historical cost measure

¹² $z' \equiv (1 - z) \left(1 + g'_{PK} \right)$ so $z' \geq 1$ as $z \geq \frac{g'_{PK}}{(1+g'_{PK})}$. For $(1 + g'_{PK}) = 1.034$ the critical value is $z^* = 0.0329$.



Appendix Figure 6.7.8 BEA Corporate Net Historical Stock Compared to IRS Book Value



Appendix Figure 6.7.9 Corporate Current-Cost Net Fixed Capital Stock Adjusted for the Great Depression and World War II

into line with the IRS book value measure and would recalibrate the other two accordingly over the interwar period. In the latter case, this is the same as multiplying the adjusted historical cost capital stock by the ratio of current- or constant-cost capital stock to historical cost over this period. Appendix figure 6.7.9 compares the official BEA 2011 measure of corporate net current-cost to the adjusted measure. The latter starts out 28% lower in 1947 and hence grows faster

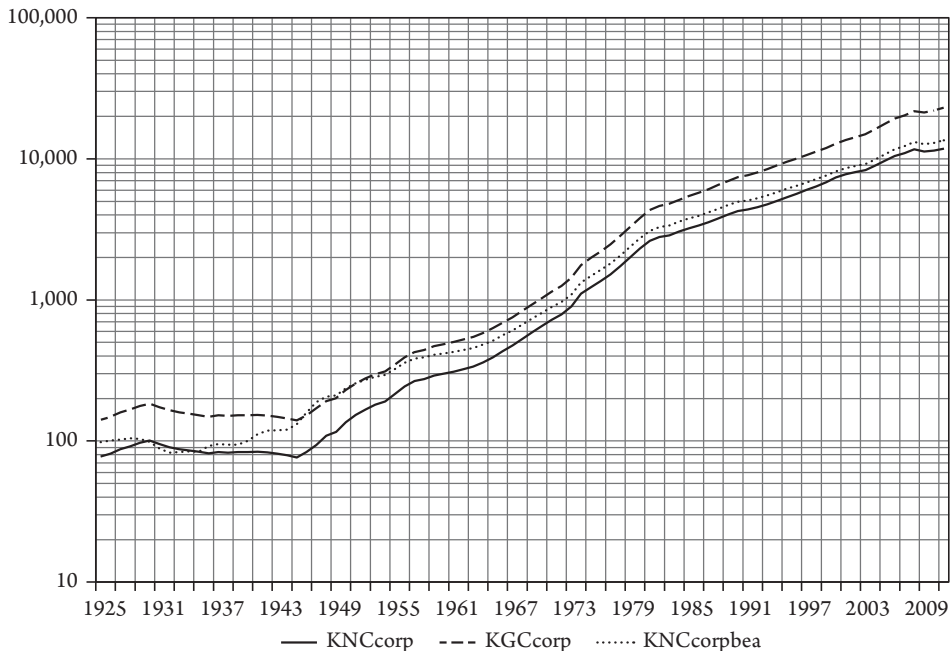
until it ends up on more or less the same path as the official measure by 1977 (see appendix table 6.8B.5).

5. Final measures of US corporate gross and net capital stocks

We can now create new estimates of gross and net capital stocks. The depletion (depreciation and retirement) rates are taken from BEA 1993. So too are the initial values for each type of stock, because these are typically estimated on the basis of assumed depletion rates. The estimated capital stocks are then adjusted in the previously described manner to allow for the effects of the Great Depression and World War II. Appendix figure 6.7.10 compares these final estimates to the corporate BEA 2011 current-cost net stock. In comparison to official BEA net capital stock (KNCcorpbea), the new net stock measure (KNCcorp) starts out lower in 1947 due to a combination of its lower initial value and the interwar effect, but then narrows the gap because it grows faster. The new gross stock measure (KNCcorp) starts out higher than the official BEA net stock, but this is more than canceled out by the interwar effect, so that by 1947 the former is 90% of the latter. However, the gross stock grows more rapidly than the official measure, so that by 2009 the former is 68% higher (see appendix table 6.8.II.5.).

6. Measures of Corporate Inventories

The remaining step is to estimate corporate inventories so as to add them to the adjusted gross current-cost capital stock. NIPA has industry data on private industries (NIPA table 5.8.5), but it is not by legal form. The Federal Reserve Board (FRB) Flow of Funds has current cost data on corporate inventories and capital stock but only for non-financial



Appendix Figure 6.7.10 Final Current-Cost Gross and Net Fixed Capital Stocks, Corporate Sector

corporations.¹³ However, the IRS publishes corporate balance sheets beginning in 1926 and these contain data on inventories, and from 1990 to 2011 also data on net historical capital stock. The IRS data is based on samples, so we cannot apply it directly to the NIPA corporate sector. The procedure therefore has two steps: first, estimation of the ratio of inventories to historical cost fixed capital for the whole period from 1947 to 2011; second, rescale the implicit inventory levels to those of the corrected capital stocks in appendix 6.8.II.5 by multiplying the preceding inventory by the ratio of adjusted historical to current-cost fixed capital stock. On the first step, the ratio of the IRS net book value stock to the BEA net historical stock in 1990–2011 turns out to be essentially a linear function of time, so it was extrapolated back to 1949–1989 and then multiplied by BEA net historical stock to yield an estimate of the corresponding IRS book value stock. This is used to construct the IRS ratio of inventories to net book value over the whole period from 1947 to 2011, which completes the first step. Multiplying this ratio by the BEA corporate net historical stock essentially scales up IRS inventory levels to match NIPA data, which completes the second step. The resulting inventory stock is, like the original IRS data, a mixture of two valuation methods: at their historical cost at the time of acquisition (First In First Out, FIFO) or at their current costs (Last In First Out, LIFO). Fixed capital (plant and equipment) is also an inventory, only of long-lived items. From that perspective, historical cost fixed capital corresponds to the FIFO methodology while current-cost fixed capital corresponds to the LIFO one. But since inventory turnover is quite rapid relative to that of fixed capital, in comparison to fixed capital FIFO inventories as valued at “fairly recent” costs while LIFO ones are at current costs. In other words, inventories are closer to current-cost than to historical cost capital stock, so we may directly add the estimated corporate inventory series to the current-cost fixed capital stock derived in appendix 6.8.II.6.

VI. Measurement of Capacity Utilization

The autoregressive distributed lag (ADL) bounds testing approach developed by Pesaran, Shin, and Smith (2001) was used to test for a long-run relationship between output and capital stock. Unlike the Engle–Granger cointegration method, the bounds testing approach is applicable regardless of whether the underlying regressors are purely $I(0)$, purely $I(1)$, mutually cointegrated or any combination of those. This is a considerable advantage given the low power of unit root tests and the relatively small size of our data for each country.

The point of departure of the bounds testing method is an error correction model (ECM) to test for the existence of a cointegrating relationship between the lagged levels of the variables. Such a connection can be interpreted as a long-run relationship. It can be assumed to exist if one rejects the F-statistic for the hypothesis that the coefficients of the lagged variable levels included in the ECM are jointly zero. Since the asymptotic distributions of both statistics are non-standard, Pesaran et al. (2001). provide two sets of asymptotic critical values for two polar cases which assume that all regressors are either purely $I(0)$ or purely $I(1)$. The critical values of all other possible cases fall between these two. If the estimated F-statistic falls outside these critical value bounds, then a conclusive inference can be drawn without any need to know about the unit root properties of the regressors. On the other hand, if the estimated statistics fall within

¹³ Non-financial inventories at current cost excluding IVA, series name = FL105015205.A; fixed capital = equipment at current cost (FL105020015.A) + residential structures at current cost (FL105012665.A) + nonresidential structures at current cost (FL105013665.A).

the critical value bounds, the inference is inconclusive and it would be necessary to establish knowledge of the unit root properties of the variables.

The ECM to be tested explains the log of output ($\ln Y$) using its own and past values of the log of the capital stock ($\ln K$) and can be written as

$$\Delta \ln Y_t = \alpha + \beta \cdot t + \sum_{h=1}^l \delta_h \cdot DM_h + \phi_1 \cdot \ln Y_{t-1} + \phi_2 \cdot \ln K_{t-1} + \sum_{i=1}^{m-1} \gamma_i \cdot \Delta \ln Y_{t-i} + \sum_{j=0}^{n-1} \varphi_j \cdot \Delta \ln K_{t-j} + \epsilon_t$$

where α is an unrestricted constant and t an unrestricted time trend that was included if its t -statistics was significant at the 5% level.¹⁴ The model can include dummy variables, DM . So as to determine which dummies to include, the squared values of the residuals were checked to identify any significantly large jumps which usually imply an “outlier” (Patterson 2000, 195). The lag structures m and n were chosen using the AIC criterion because it tends to select a greater numbers of lags than the BIC and a “sufficient” number of lags is required eliminate serial correlation.¹⁵

In order to test for the existence of a long-run relationship between $\ln Y$ and $\ln K$, the ECM was estimated via OLS and the null hypotheses was tested that $H_0 : \phi_1 = \phi_2 = 0$, using tables CI(iv) and CI(iii) in Pesaran (2001, 300–301) for the model with and without a trend.¹⁶ Rejection of the null hypothesis allows one to conclude that a long-run relationship exists. Since the bounds test requires the absence of serial correlation in the ECM, the data was previously tested for autocorrelation using the Breusch–Godfrey LM test.¹⁷ In cases where the bounds F -test statistic fell between the two critical values and implied an inconclusive result, we followed the Engle–Granger two-step procedure, investigated the order of integration of both time series and tested for the existence of a cointegrating relationship between them with Dickey–Fuller tests.

Wherever it was possible to identify a cointegrating long-term relationship, we used the corresponding ADL(m, n) model

$$\ln Y_t = \alpha + \beta \cdot t + \sum_{h=1}^l \delta_h \cdot DM_h + \sum_{i=1}^m \gamma_i \cdot \ln Y_{t-i} + \sum_{j=0}^{n-1} \varphi_j \cdot \ln K_{t-j} + \epsilon_t$$

to obtain the coefficients of the long-run relation

$$\ln Y^* = a + b \cdot t + c_h \cdot DM_h + d \cdot \ln K^* + \epsilon_t$$

where $a = \frac{\alpha}{1 - \sum \gamma_i}$, $b = \frac{\beta}{1 - \sum \gamma_i}$, $c_h = \frac{\delta_h}{1 - \sum \gamma_i}$ and $d = \frac{\sum \varphi_j}{1 - \sum \gamma_i}$. The coefficients $\alpha, \beta, \delta_h, \gamma_i$, and φ_j were saved from the OLS estimation of the ADL(m, n) model and used to calculate a, b, c_h and d .

¹⁴ This was tested in the very first step in an ECM with a time trend and (arbitrarily) four lags for the differences of both variables (which is the ECM that corresponds to an ADL(5,5) model).

¹⁵ As this is commonly done by others in the particular application here, we tested all combinations of lags from t to $t - i$ and t to $t - j$ with $i \leq 5$ and $j \leq 5$.

¹⁶ Due to its low power in the present context, we omitted running t -tests on the lagged log of the capital stock.

¹⁷ We increased the number of the lags and/or altered the use of dummies whenever serial correlation was detected.

Appendix Table 6.7.14 Capacity Utilization Regression Output

Source	SS	df	MS	Number of obs	=	61
				F(10, 50)	=	6256.98
Model	16.5869448	10	1.65869448	Prob > F	=	0
Residual	0.01325476	50	0.0002651	R-squared	=	0.9992
				Adj R-squared	=	0.999
Total	16.6001995	60	0.27666999	Root MSE	=	0.01628

<i>Coefficient Estimates</i>						
l_ys	Coef.	Std. Err.	T	P>t	[95% Conf.	Interval]
l_ys						
L1.	1.298062	0.0967548	13.42	0	1.103724	1.492399
L2.	-0.3930554	0.0961629	-4.09	0	-0.5862043	-0.1999065
l_ks						
-.	5.085575	0.5376103	9.46	0	4.005753	6.165397
L1.	-14.28483	1.21512	-11.76	0	-16.72547	-11.84419
L2.	15.70543	1.735902	9.05	0	12.21877	19.19209
L3.	-8.296103	1.478974	-5.61	0	-11.26671	-5.325497
L4.	1.852707	0.5527994	3.35	0.002	0.7423769	2.963037
d74	-0.0811995	0.0170822	-4.75	0	-0.1155101	-0.0468889
d56	-0.0705601	0.0170684	-4.13	0	-0.1048431	-0.0362772
d80	-0.0454098	0.0173955	-2.61	0.012	-0.0803496	-0.01047
_cons	0.2069159	0.0638755	3.24	0.002	0.0786181	0.3352137

<i>Information criteria</i>						
Model	Obs	ll(null)	ll(model)	df	AIC	BIC
	61	-46.86075	170.6901	11	-319.3801	-296.1605

<i>Long run coefficients</i>	
$a = \alpha / (1 - \Sigma \gamma)$	2.1782063
$b = \beta / (1 - \Sigma \gamma)$	0
$c1 = \delta 1 / (1 - \Sigma \gamma)$	-0.8547887
$c2 = \delta 2 / (1 - \Sigma \gamma)$	-0.7427876
$c3 = \delta 3 / (1 - \Sigma \gamma)$	-0.4780294
$d = \Sigma \varphi / (1 - \Sigma \gamma)$	0.66088568

The final step was to estimate $\ln Y^*$, which we interpret as the log of capacity ($\ln U$) by applying the imputed long-run relationship coefficients on the capital stock time series (and the dummies and time dimension). The cointegration-based estimate of capacity utilization was then obtained as $u = e^{\ln U}$, as previously depicted in appendix figure 6.6.1. Table 6.7.14 summarized the regression output. A comparison between new measure and the Federal Reserve Board measure (available from 1967) is presented in chapter 6, figure 6.4.

VII. Final Measures of Profit, Capital, and the Rate of Profit

Appendix 6.7, section I.2 established that the profit rate of the non-corporate sector (corrected for the wage-equivalent of proprietors and partners) is close to the rate of the corporate sector. We are therefore justified in treating the corporate sector as representative. Four new variables are relevant here: (1) corporate value added adjusted for imputed net interest; (2) corporate net operating surplus adjusted for imputed net interest which makes it the same as corporate profit inclusive of net monetary interest paid or what businesses call Earnings before Interest and Taxes (EBIT); (3) gross corporate current-cost capital stock adjusted for the effects of the interwar period; and (4) the rate of capacity utilization which is the ratio of actual output to normal-capacity output. The ratio of the output to capital defines the maximum rate of profit (output–capital ratio) and that of profit to capital the average rate of profit. The corresponding BEA measures are in terms of official corporate value added (inclusive of imputed net interest) and corporate profit, relative to the official BEA corporate net capital stock. Each pair must be further adjusted for capacity utilization.

Chapter 6, section VIII, reviews the derivation of the new output, value added, profit, capital, and capacity utilization measures developed in this appendix, presents and analyzes the resulting profitability patterns, and compares them to conventional measures. Table 6.22 in chapter 6 summarizes the differences between the 1947–1982 “golden-age” and the 1982–2011 neoliberal eras. Finally, it is noted that once the GPIM rules are understood, new measures of capital stock and capacity can easily be derived even at the industry level, and these two factors turn out to be sufficient to develop good proxies for the corrected measures even without adjustment for imputed interest or inventories (chapter 6, figure 6.6). Even more strikingly, it is found that the incremental rate of profit (the rate of return on new investment) which is so critical to the classical theory of inter-sectoral profit rate equalization can be approximated to a high degree without having to make adjustments for imputed interest, inventories or even capital stock retirement patterns (chapter 6, figure 6.7).